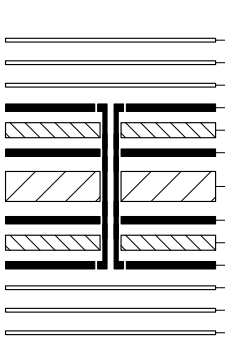
BMS_MASTER Fabrication Document

Layer Stack Legend



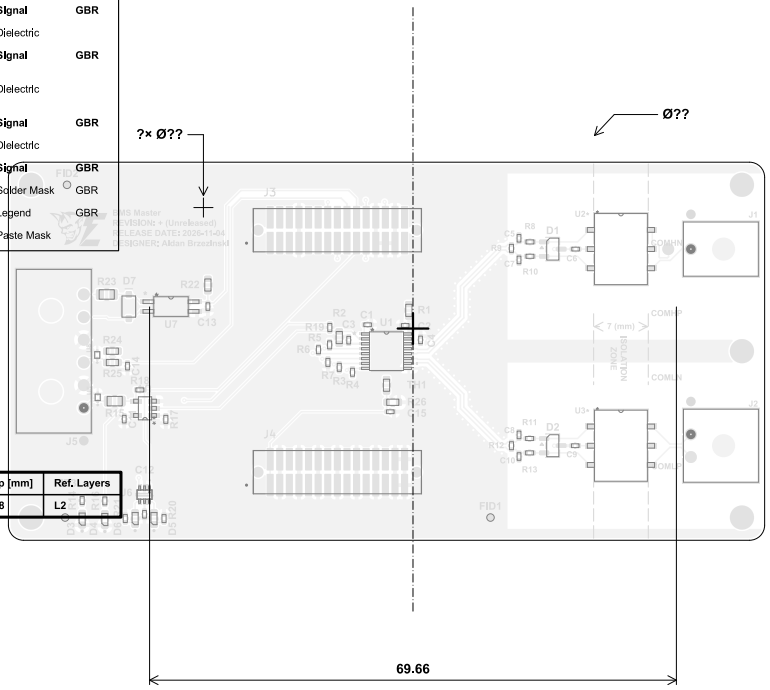
Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen		Direct Printing	Legend	GBR
	F.Mask	0.02mm	Solder Resist	Solder Mask	GBR
Copper	**L1**	**0.035mm (1.00oz)**		**Signal**	**GBR**
Prepreg		0.1855mm	FR4_7628	Dielectric	
Copper	**L2**	**0.035mm (1.00oz)**		**Signal**	**GBR**
Core		1.03mm	FR4	Dielectric	
Copper	**L3**	**0.035mm (1.00oz)**		**Signal**	**GBR**
Prepreg		0.1855mm	FR4_7628	Dielectric	
Copper	**L4**	**0.035mm (1.00oz)**		**Signal**	**GBR**
	B.Mask	0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen		Direct Printing	Legend	GBR
	B.Paste			Paste Mask	

Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0.2032	0.28	L2

Top Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

4) SURFACE FINISH: IMMERSION GOLD

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLUE.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY BLUE DEVIL MOTOR SPORTS.

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	100.000 × 50.000 mm
BOARD THICKNESS	1.581 mm
TRACE WIDTH	0.200 mm
TRACE TO TRACE	-0.000 mm
MIN. HOLE (PTH)	0.250 mm
MIN. HOLE (NPTH)	1.400 mm
ANNULAR RING	0.150 mm
COPPER TO HOLE	0.254 mm
COPPER TO EDGE	0.500 mm
HOLE TO HOLE	0.254 mm

11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.

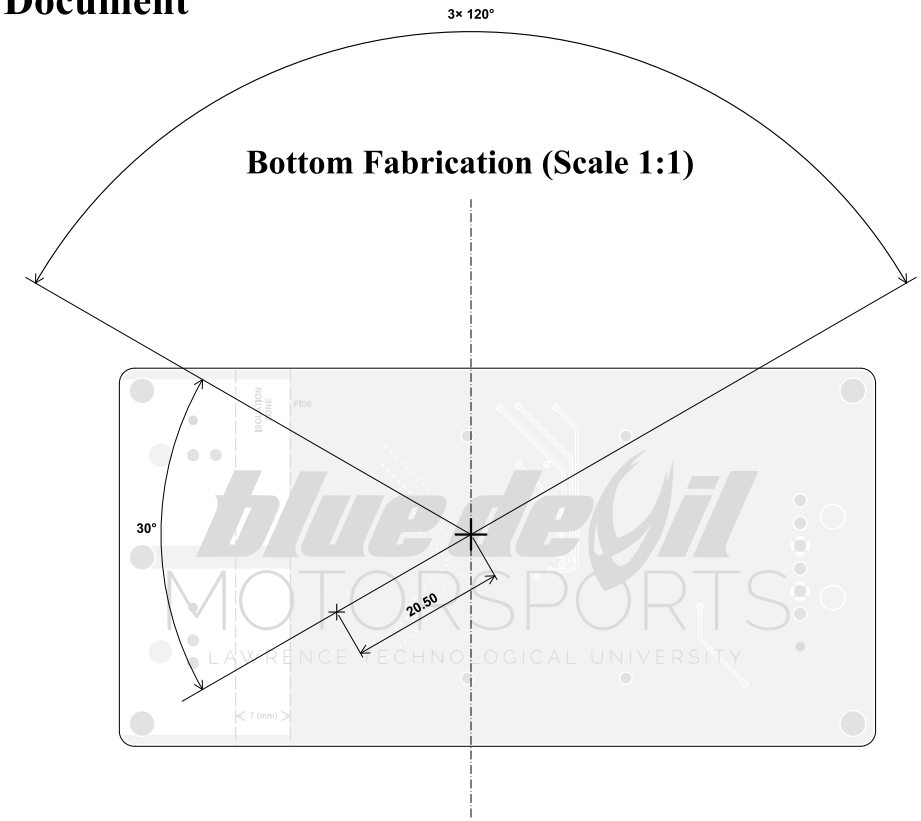
12) CONFIRM SPACE WIDTHS AND SPACINGS.

blue devil MOTORSPORTS

Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: 17caac0
	Board Name: **BMS_MASTER**		Project Name: **BMS Master**	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: bms_master.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **1** of **10**

BMS_MASTER Fabrication Document

Bottom Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

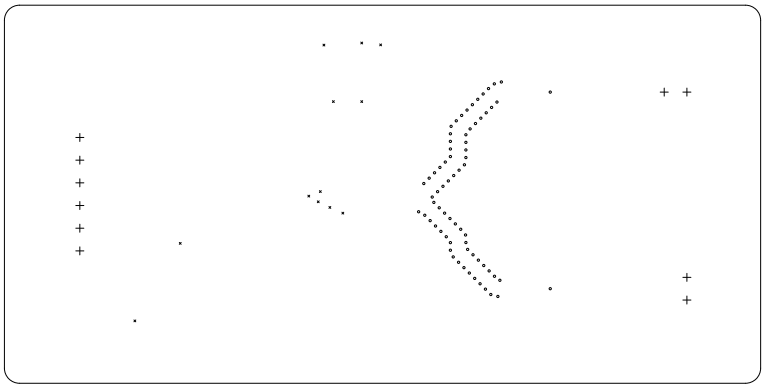
Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: 17caac0
	Board Name: BMS_MASTER		Project Name: BMS Master	
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: bms_master.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 2 of 10

BMS_MASTER Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	12	0.25mm (9.84mils)	PTH	Round	L1 - L4	Via
○	70	0.30mm (11.81mils)	PTH	Round	L1 - L4	Via
+	10	1.07mm (42.00mils)	PTH	Round	L1 - L4	Pad
	Total 92					

Drill Drawing L1 - L4 (Scale 1:1)



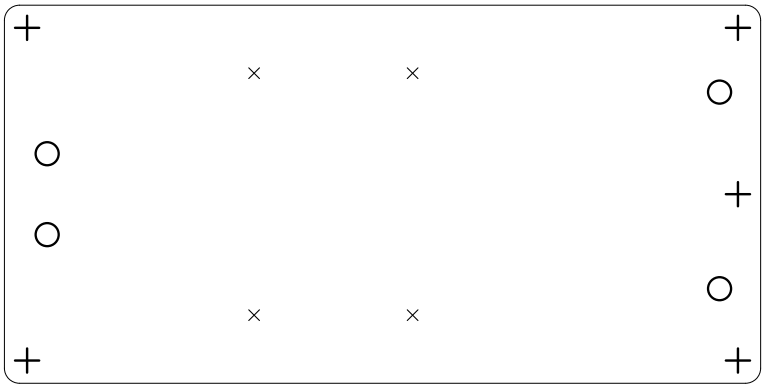
Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: 17caac0	
	Board Name: BMS_MASTER		Project Name: BMS Master		
Sheet Title: Drill Drawing (L1 - L4)	File Name: bms_master.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)	
Sheet Path:		Reviewer:	Size: A4	Sheet: 3 of 10	

BMS_MASTER Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	4	1.42mm (55.91mils)	NPTH	Round	L1 - L4	Mechanical
○	4	3.05mm (120.00mils)	NPTH	Round	L1 - L4	Mechanical
+	5	3.20mm (125.98mils)	NPTH	Round	L1 - L4	Mechanical
	Total 13					

Drill Drawing L1 - L4 (Scale 1:1)



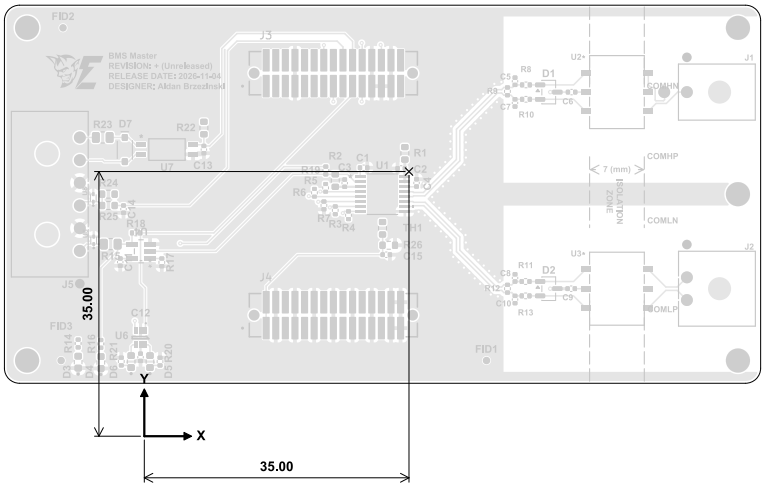
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	Board Name: BMS_MASTER		Project Name: BMS Master		
Sheet Title: Drill Drawing (L1 - L4)	File Name: bms_master.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)	
Sheet Path:		Reviewer:	Size: A4	Sheet: 4 of 10	

BMS_MASTER Fabrication Document

Top Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]

Ref.	Net	X [mm]	Y [mm]



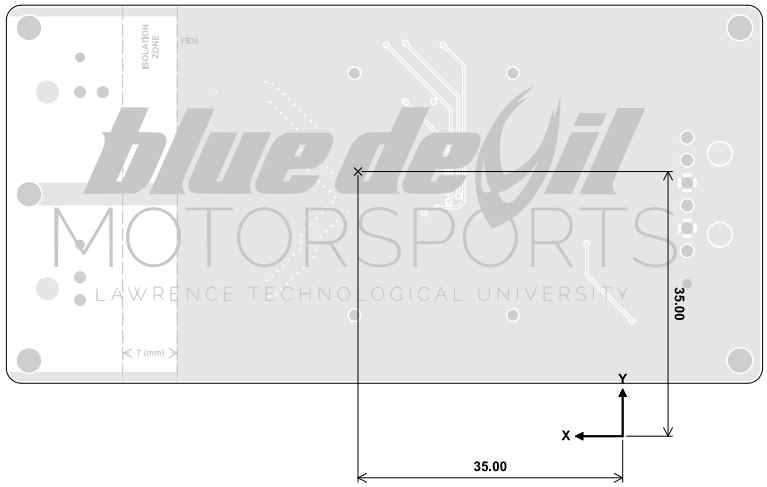
All dimensions are in millimeters unless otherwise specified.

Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: 17caac0
	Board Name: BMS_MASTER		Project Name: BMS Master	
Sheet Title: Top Test Points (Scale 1:1)	File Name: bms_master.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 5 of 10

BMS_MASTER Fabrication Document

Bottom Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]



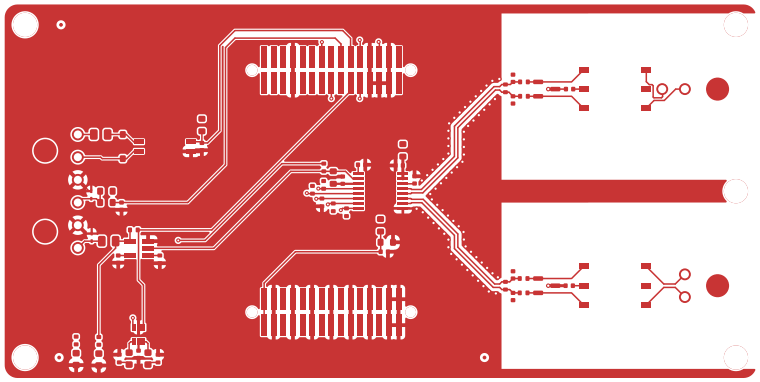
blue devil MOTORSPORTS
LAWRENCE TECHNOLOGICAL UNIVERSITY

All dimensions are in millimeters unless otherwise specified.

Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: 17caac0
	Board Name: BMS_MASTER		Project Name: BMS Master	
Sheet Title: Bottom Test Points (Scale 1:1)	File Name: bms_master.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 6 of 10

BMS_MASTER Fabrication Document

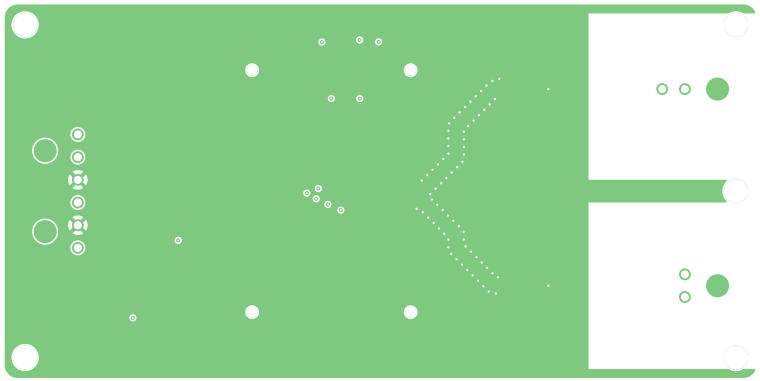
L1 (Scale 1:1)



Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: 17caac0
	Board Name: BMS_MASTER		Project Name: BMS Master	
Sheet Title: L1 (Scale 1:1)	File Name: bms_master.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 7 of 10

BMS_MASTER Fabrication Document

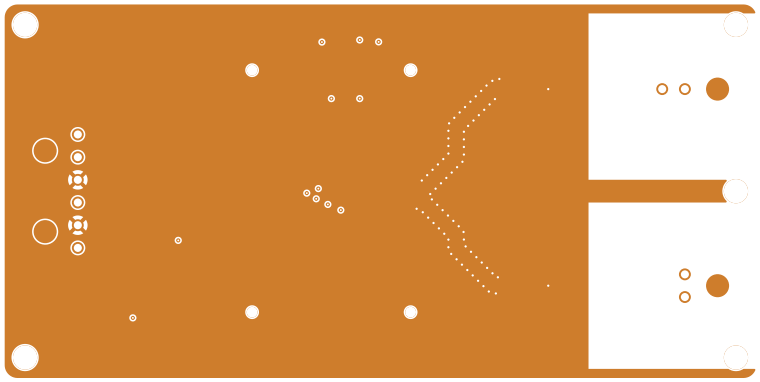
L2 (Scale 1:1)



Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: 17caac0
	Board Name: BMS_MASTER		Project Name: BMS Master	
Sheet Title: L2 (Scale 1:1)	File Name: bms_master.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 8 of 10

BMS_MASTER Fabrication Document

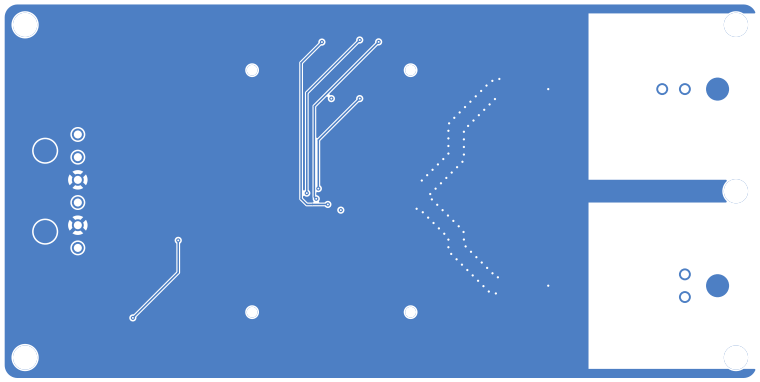
L3 (Scale 1:1)



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	Board Name: BMS_MASTER		Project Name: BMS Master	
Sheet Title: L3 (Scale 1:1)	File Name: bms_master.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 9 of 10

BMS_MASTER Fabrication Document

L4 (Scale 1:1)



Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: 17caac0
	Board Name: BMS_MASTER		Project Name: BMS Master	
Sheet Title: L4 (Scale 1:1)	File Name: bms_master.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 10 of 10